

Dead Beat Control of Three Phase PWM Inverter

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Abstract—A new method of the microprocessor control of three phase sinusoidal voltage pulsewidth modulated (PWM) inverters is proposed. First, the discretized state equations of the inverter main circuit on the d - q frame have been derived. An algorithm of the dead beat control with a current minor loop that constrains the inverter current within the safety limit, is subsequently developed. To compensate the computing time delay, the second order prediction method, and a new discretization method using twice time of the sampling period have been adopted. This method is especially suitable for inverters using high speed switching devices and digital signal processors.

I. INTRODUCTION

AMONG many kinds of converters, the PWM inverter has most applications such as uninterruptible power systems (UPS's), active filters, high power factor converters, and adjustable frequency drives. Recently, according to the progress of high speed switching devices and digital controllers such as DSP, a new development stage of the PWM inverter can be expected.

Almost all applications of PWM inverters except UPS have the controller structure with a current minor loop, and performance of the inverter system largely depends on the quality of the current minor loop. Therefore current control of PWM inverter is one of the most important subject of power electronics.

In [1], the authors have proposed a new single phase UPS with a dead beat current minor loop and a voltage major loop, implemented with two 8-bit microprocessors. The authors have also presented a high performance current minor loop using high speed switching devices and the dead beat control. In addition, feasibility of UPS having a current minor loop was also confirmed. Then it became possible to unify almost all applications of PWM inverters including UPS with the current minor loop.

Based on the above mentioned background, the authors have been trying to extend technology of the dead beat current minor loop to three phase inverters. This paper presents a principle of three phase sinusoidal voltage inverter with a current minor loop, method of constructing the dead beat current and voltage controllers, and results of simulation using a hybrid computer.

II. PRINCIPLE OF SINUSOIDAL VOLTAGE INVERTER

Principle of proposed three phase sinusoidal inverter with a current minor loop is shown in Fig. 1. As the typ-

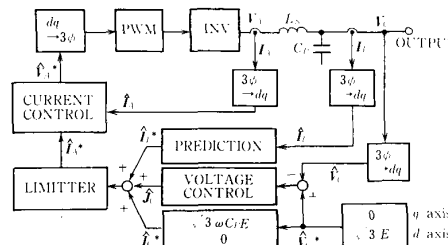


Fig. 1. Sinusoidal voltage inverter with current minor loop. (^ Marks used for variables on d - q frame.)

ical inverter main circuit, the three phase bridge inverter is supposed, but this principle can be used generally for other types of three phase voltage inverters.

To obtain high performance of control, the controller is constructed on the synchronizing d - q frame. Three phase inverter variables such as V_C and I_A are transformed to variables on the d - q frame such as $\hat{V}_C$ and $\hat{I}_A$. For simplicity the PWM carrier is synchronized to the sampling of the digital control. When switching devices can operate at higher switching frequency f_{sw} than the sampling frequency f_s , the PWM frequency is selected to be $f_{sw} = n \cdot f_s$ (n = integer) as shown in Fig. 2. This means that same pulsewidth is output two or three times in one sampling period. By doing so, the inverter can switch beyond the audible frequency and the ripple current can also be decreased, even if the sampling frequency of DDC is lower than the upper limit of audible frequency.

Various methods of current control can be used for this system, but in this paper the dead beat control is adopted as explained below. By using the dead beat control, current response of one discretization time can be obtained.

As the reference of the current minor loop, sum of following three components is given through a limiter, as shown in Fig. 1.

First, the current reference $\hat{I}_C^*$ to flow through the parallel filter capacitance C_p is given. No load voltage will be established by giving a q -axis current reference $\hat{I}_{cq}^* = \sqrt{3} \omega C_p E$ as the current for the parallel filter capacitance C_p .

Secondly, the feed-forward of the load current $\hat{I}_L^*$ is given. A predicted value of the load current after the discretization time T is obtained using the second order prediction method on the d - q frame. The current minor loop follows this feed-forward signal and responds to distortions and disturbances of load current.

Thirdly, the current command $\hat{I}_C$ from the voltage major loop is given. The voltage controller stabilizes the sys-

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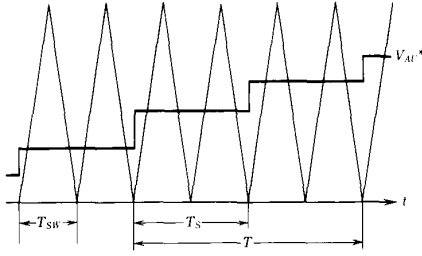


Fig. 2. Relation of switching carrier, DDC signal and discretization time (when $n = 2$).

tem and assures constant voltage characteristic, by correcting instantaneous error of the output voltage caused by following factors:

- too fast change of the load current that the current minor loop cannot follow
- sudden change of di/dt of the load current that causes error of prediction
- error of the current loop caused by deviation of circuit constant L_s , C_p from designed value
- error of the current loop caused by switching delay time T_d of the inverter.

If the output voltage deviates by above factors, the load current changes from original wave form, so that the current loop follows this deviated load current through prediction, and results in an unstable system. To stabilize the system against above mentioned factors, the voltage major loop also should have quick response. For that reason the dead beat control of proportional type is adopted for the voltage major loop, and the authors call this system the double dead beat control system.

Providing a limiter for the reference $\hat{I}_A^*$ of the current minor loop, an easy applicable sinusoidal voltage inverter with inherent over current protection can be obtained.

III. DECOUPLED DOUBLE DEAD BEAT CONTROL SYSTEM

A. State Equation of Three Phase Expression

The main circuit of the three phase inverter is shown in Fig. 3.

In this figure and following equations, meaning of variables and suffixes are as follows:

V_A	inverter output terminal voltage from neutral
V_C	condenser voltage from V_X
I_A	inverter current
I_C	condenser current
I_L	load current
V_X	condenser neutral voltage from neutral
U, V, W	components of three phase expression
d, q	components of d - q frame expression
*	command value
ω	angular velocity of output frequency
T	discretization time
$\wedge$	vector expression of variables on q - d -0 or q - d frame.

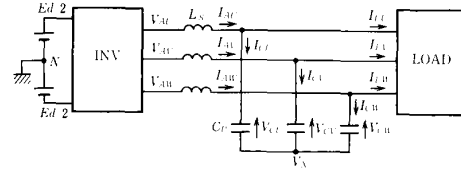


Fig. 3. Main circuit.

Using the vector equation, the main circuit of Fig. 3 can be expressed by

$$C_p \frac{d}{dt} V_C = I_A - I_L \quad (1)$$

$$L_s \frac{d}{dt} I_A = V_A - V_C - V_X \quad (2)$$

where each variable such as I_A , V_C , etc., are vector variables given by (3) with suffixes of each variable. L_s and C_p are given by (4), and V_X is given by following (5):

$$I = \text{col}[I_U, I_V, I_W] \quad (3)$$

$$V = \text{col}[V_U, V_V, V_W]$$

$$L_s = \text{diag}[L_s, L_s, L_s] \quad (4)$$

$$C_p = \text{diag}[C_p, C_p, C_p]$$

$$V_X = (V_{AU} + V_{AV} + V_{AW})/3. \quad (5)$$

From (2) and (5)

$$L_s \frac{d}{dt} I_A = DV_A - V_C \quad (6)$$

where

$$D = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}. \quad (7)$$

B. State Equation on Synchronizing d - q Frame

Equations (1) and (6) will be transformed to the q - d -0 frame by using the following transformation matrix C , in relation to (9), [6].

$$C = \sqrt{\frac{3}{2}} \begin{pmatrix} \cos \omega t & \cos (\omega t - 2\pi/3) & \cos (\omega t + 2\pi/3) \\ \sin \omega t & \sin (\omega t - 2\pi/3) & \sin (\omega t + 2\pi/3) \\ 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}. \quad (8)$$

$$\begin{aligned} \hat{V} &= \text{col}[V_q, V_d, V_0] \\ &= C \text{col}[V_U, V_V, V_W] = CV \end{aligned} \quad (9)$$

$$\begin{aligned} \hat{I} &= \text{col}[I_q, I_d, I_0] \\ &= C \text{col}[I_U, I_V, I_W] = CI \end{aligned}$$

As the result of transformation, (6) reduces to (10).

$$\begin{pmatrix} 0 & \omega L_S & 0 \\ -\omega L_S & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \hat{\mathbf{I}}_A + L_S \frac{d}{dt} \hat{\mathbf{I}}_A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} (\hat{\mathbf{V}}_A - \hat{\mathbf{V}}_C) \quad (10)$$

Since 0-axis components has no relation to other axes, these components can be omitted from (10). Then state equations (11) of the inverter current can be derived:

$$\begin{pmatrix} \dot{I}_{Aq} \\ \dot{I}_{Ad} \end{pmatrix} = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix} \begin{pmatrix} I_{Aq} \\ I_{Ad} \end{pmatrix} + \begin{pmatrix} 1/L_S & 0 \\ 0 & 1/L_S \end{pmatrix} \begin{pmatrix} V_{Aq} \\ V_{Ad} \end{pmatrix} - \begin{pmatrix} 1/L_S & 0 \\ 0 & 1/L_S \end{pmatrix} \begin{pmatrix} V_{cq} \\ V_{cd} \end{pmatrix}. \quad (11)$$

Next, as the result of transformation of the condenser voltage equation (1), the following equation can be derived:

$$\begin{pmatrix} 0 & \omega C_P & 0 \\ -\omega C_P & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \hat{\mathbf{V}}_C + C_P \frac{d}{dt} \hat{\mathbf{V}}_C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} (\hat{\mathbf{I}}_A - \hat{\mathbf{I}}_L). \quad (12)$$

Also in this equation, 0-axis components can be omitted, and the state equation (13) of the condenser voltage can be derived:

$$\begin{pmatrix} \dot{V}_{cq} \\ \dot{V}_{cd} \end{pmatrix} = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix} \begin{pmatrix} V_{cq} \\ V_{cd} \end{pmatrix} + \begin{pmatrix} 1/C_P & 0 \\ 0 & 1/C_P \end{pmatrix} \begin{pmatrix} I_{Aq} \\ I_{Ad} \end{pmatrix} - \begin{pmatrix} 1/C_P & 0 \\ 0 & 1/C_P \end{pmatrix} \begin{pmatrix} I_{Lq} \\ I_{Ld} \end{pmatrix}. \quad (13)$$

Reference of the output voltage and the condenser current on the d - q frame will be given as follows.

First, the output voltage reference of (14) is transformed by matrix $\mathbf{C}$ as (15), and the reference value $\sqrt{3}E$ on the d -axis is obtained:

$$\mathbf{V}_C^* = \begin{pmatrix} V_{CU}^* \\ V_{CQ}^* \\ V_{CW}^* \end{pmatrix} = \begin{pmatrix} \sqrt{2}E \sin(\omega t) \\ \sqrt{2}E \sin(\omega t - 2\pi/3) \\ \sqrt{2}E \sin(\omega t + 2\pi/3) \end{pmatrix} \quad (14)$$

$$\hat{\mathbf{V}}_C^* = \begin{pmatrix} V_{Cq}^* \\ V_{Cd}^* \end{pmatrix} = \mathbf{C} \mathbf{V}_C^* = \begin{pmatrix} 0 \\ \sqrt{3}E \end{pmatrix}. \quad (15)$$

Equation (16) of the condenser current reference by three phase expression is transformed to the d - q frame as

(17):

$$\mathbf{I}_C^* = \mathbf{C}_P \frac{d}{dt} \mathbf{V}_C^* \quad (16)$$

$$\begin{aligned} \hat{\mathbf{I}}_C^* &= \begin{pmatrix} I_{Cq}^* \\ I_{Cd}^* \end{pmatrix} = \mathbf{C} \mathbf{I}_C^* = \mathbf{C} \mathbf{C}_P \frac{d}{dt} \mathbf{V}_C^* = \mathbf{C} \mathbf{C}_P \frac{d}{dt} (\mathbf{C}^{-1} \hat{\mathbf{V}}_C^*) \\ &= \left(\mathbf{C} \mathbf{C}_P \frac{d}{dt} \mathbf{C}^{-1} \right) \mathbf{V}_C^* + \mathbf{C} \mathbf{C}_P \mathbf{C}^{-1} \frac{d}{dt} \mathbf{V}_C^* \\ &= \begin{pmatrix} 0 & \omega C_P \\ -\omega C_P & 0 \end{pmatrix} \hat{\mathbf{V}}_C^* = \begin{pmatrix} \sqrt{3}\omega C_P E \\ 0 \end{pmatrix}. \end{aligned} \quad (17)$$

C. Discretization of the State Equations

Here state equations (11) and (13) of the inverter current and the voltage will be discretized separately. Separate two step discretization of the current equation (11) and the voltage equation (13) means

- that the condenser voltage $\hat{\mathbf{V}}_C$ is assumed to be constant during the discretization time T of the inverter current, and
- that the inverter is assumed to be an ideal current source and its current $\hat{\mathbf{I}}_A$ is assumed to be constant during the discretization time T of the condenser voltage.

This method is simple and has sufficient accuracy when $\sin \omega T \approx \omega T$ is valid.

First, (11) is rewritten as (18), and matrix $\mathbf{A}$ and $\mathbf{F}$ of its discretized state equation (19) are derived:

$$\hat{\mathbf{I}}_A(t) = \mathbf{A}' \hat{\mathbf{I}}_A(t) + \mathbf{F}' \hat{\mathbf{V}}_A(t) - \mathbf{F}' \hat{\mathbf{V}}_C(t) \quad (18)$$

$$\hat{\mathbf{I}}_A(k+1) = \mathbf{A} \hat{\mathbf{I}}_A(k) + \mathbf{F} \hat{\mathbf{V}}_A(k) - \mathbf{F} \hat{\mathbf{V}}_C(k). \quad (19)$$

Next, (13) is rewritten as (20), and matrix $\mathbf{K}$ of its discretized form (21) is derived:

$$\hat{\mathbf{V}}_C(t) = \mathbf{A}' \hat{\mathbf{V}}_C(t) + \mathbf{K}' \hat{\mathbf{I}}_A(t) - \mathbf{K}' \hat{\mathbf{I}}_L(t) \quad (20)$$

$$\hat{\mathbf{V}}_C(k+1) = \mathbf{A} \hat{\mathbf{V}}_C(k) + \mathbf{K} \hat{\mathbf{I}}_A(k) - \mathbf{K} \hat{\mathbf{I}}_L(k). \quad (21)$$

The method and result of the previously mentioned matrix calculation are as follows [5]:

$$\mathbf{A} = L^{-1}[(s\mathbf{I} - \mathbf{A}')^{-1}] = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \quad \mathbf{F} = \int_0^T \mathbf{A}(T-\tau) \mathbf{F}' d\tau = \begin{pmatrix} c & -d \\ d & c \end{pmatrix} \quad (22)$$

$$\mathbf{K} = \int_0^T \mathbf{A}(T-\tau) \mathbf{K}' d\tau = \begin{pmatrix} e & -f \\ f & e \end{pmatrix} \quad a = \cos \omega T, \quad b = \sin \omega T, \quad c = \frac{1}{\omega L_S} \sin \omega T$$

$$d = \frac{2}{\omega L_S} \sin^2 \frac{\omega T}{2}, \quad e = \frac{1}{\omega C_P} \sin \omega T \quad (23)$$

$$f = \frac{2}{\omega C_P} \sin^2 \frac{\omega T}{2}$$

D. Decoupled Dead Beat Current Controller

Equation (19) can be Z transformed as following (24):

$$Z\hat{I}_A(Z) = A\hat{I}_A(Z) + F\hat{V}_A(Z) - F\hat{V}_C(Z). \quad (24)$$

This equation can be expressed by a block diagram shown in the right side of the dotted line of the Fig. 4.

First, to compensate the disturbance term of $\hat{V}_C(Z)$, the condenser voltage is detected and added to the output of the controller as the counter electromotive force (EMF) compensation signal.

Secondly, to decouple the coupling elements d of the coefficient matrix F of the control variable $\hat{V}_A(Z)$, a decoupling matrix F_{dc} is provided.

Here, F_{dc} should satisfy the following equation:

$$FF_{dc} = c; \quad (25)$$

then,

$$\begin{aligned} F_{dc} &= F^{-1}c = \frac{c}{c^2 + d^2} \begin{pmatrix} c & d \\ -d & c \end{pmatrix} \\ &= \frac{1}{c[1 + (\tan \omega T/2)^2]} \begin{pmatrix} c & d \\ -d & c \end{pmatrix} \\ &\approx \begin{pmatrix} 1 & d/c \\ -d/c & 1 \end{pmatrix}. \end{aligned} \quad (26)$$

Where, usually in case of high sampling rate,

$$(d/c)^2 = (\tan \omega T/2)^2 \approx (\omega T/2)^2 \ll 1. \quad (27)$$

Next, to decouple the coupling elements b of the matrix A , feedback through a decoupling matrix A_{dc1} is provided.

As can be seen from Fig. 4 and (22), $FF_{dc}A_{dc1}$ should be opposite polarity of the coupling elements of the matrix A . Therefore we obtain the following relation:

$$FF_{dc}A_{dc1} = \begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix}. \quad (28)$$

And using (25),

$$A_{dc1} = \begin{pmatrix} 0 & b/c \\ -b/c & 0 \end{pmatrix}. \quad (29)$$

After removing couplings as mentioned above, then the dead beat controller of I - P function is provided as shown in the Fig. 4.

Adopting the I - P control, influence of the switching dead time T_d can be compensated quickly when the inverter current crosses zero.

To get finite time settlement response, the condition necessary to nullify two roots of the closed loop transfer function of the block diagram of Fig. 4 is derived as

$$G_1 = 1/c, \quad G_2 = a/c. \quad (30)$$

As the result of above mentioned construction, the transfer function of the current minor loop becomes as follows:

$$\hat{I}_A = (1/Z)\hat{I}_A^*. \quad (31)$$

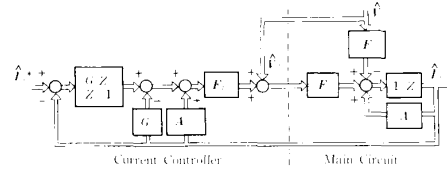


Fig. 4. Decoupled dead beat current control system.

E. Decoupled Dead Beat Voltage Controller

Equation (21) can be Z transformed as following (32):

$$Z\hat{V}_C(Z) = A\hat{V}_C(Z) + K\hat{I}_A(Z) - K\hat{I}_L(Z). \quad (32)$$

This equation can be expressed by a block diagram shown in the right side of the dotted line of the Fig. 5.

In this figure, the current minor loop of the Fig. 4 is assumed to be a ideal current source, and shown by a block current minor loop (CML).

From this figure, it can be seen that the inverter must quickly supply the load current $\hat{I}_L$ to compensate the disturbance of the load.

Therefore, the block prediction (PRD) for the prediction that will be explained in Section IV has been adopted.

Using the predicted load current as the target value of the current minor loop, the delay of the current minor loop can be compensated.

Next, to decouple matrices K and A , matrices K_{dc} and A_{dc2} given by (33) and (34), are provided using same method as mentioned in the current controller:

$$\begin{aligned} K_{dc} &= K^{-1}e = \frac{e}{e^2 + f^2} \begin{pmatrix} e & f \\ -f & e \end{pmatrix} \\ &\approx \begin{pmatrix} 1 & f/e \\ -f/e & 1 \end{pmatrix} \end{aligned} \quad (33)$$

$$A_{dc2} = \begin{pmatrix} 0 & b/e \\ -b/e & 0 \end{pmatrix}. \quad (34)$$

After removing couplings as mentioned above, the dead beat voltage controller of proportional function is provided as shown in Fig. 5.

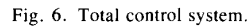
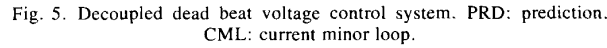
Condition for the finite time settlement response of this voltage controller is

$$G_3 = a/e. \quad (35)$$

As for the condenser current, it can be previously known from the output voltage reference using (17). Considering the recovery of the output voltage from nonlinear operation caused by big disturbances, it is better to give this condenser current $I_{Cq}^* = \sqrt{3}\omega C_p E$ as a feedforward value. The signal that corresponds to this condenser current is included in the decoupling term $A_{dc2}\hat{V}_C$. Then this signal is separated to two components as

$$A_{dc2}\hat{V}_C = A_{dc2}\hat{V}_C^* - A_{dc2}(\hat{V}_C^* - \hat{V}_C) \quad (36)$$

The first term $A_{dc2}\hat{V}_C^*$ is multiplied by K_{dc} , and the adding point is moved after the decoupling matrix K_{dc} . This


$$\begin{aligned} \mathbf{K}_{\text{dc}} \mathbf{A}_{\text{dc}2} \hat{\mathbf{V}}_C^* &= \begin{pmatrix} 1 & f/e \\ -f/e & 1 \end{pmatrix} \begin{pmatrix} 0 & b/e \\ -b/e & 0 \end{pmatrix} \begin{pmatrix} V_{Cq}^* \\ V_{Cd}^* \end{pmatrix} \\ &= \frac{b}{e} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} V_{Cq}^* \\ V_{Cd}^* \end{pmatrix} + \frac{bf}{e^2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \\ &\quad \cdot \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} V_{Cq}^* \\ V_{Cd}^* \end{pmatrix} \\ &\cong \omega C_P \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} V_{Cq}^* \\ V_{Cd}^* \end{pmatrix} = \begin{pmatrix} \sqrt{3} \omega C_P E \\ 0 \end{pmatrix}. \end{aligned} \quad (37)$$

IV. PREDICTION OF LOAD CURRENT AND COMPENSATION OF COMPUTING DELAY

Another important problem is compensation of computing time delay and full utilization of the inverter pulse-width to enable output of the inverter voltage while the

In this paper, the second order prediction method using the following equation will be introduced:

$$\hat{I}_L^*(m) = 6\hat{I}_L(m) - 8\hat{I}_L(m-1) + 3\hat{I}_L(m-2). \quad (38)$$

This formula gives value of $\hat{I}_L^*(m)$ on the quadratic curve which goes through $\hat{I}_L(m-2)$, $\hat{I}_L(m-1)$ and $\hat{I}_L(m)$ at $t = (m-2)T_s$, $(m-1)T_s$ and mT_s , respectively. To realize load current $\hat{I}_L^*(m)$ shown by a mark $\bigcirc$ in Fig. 7, after the discretization time $T = 2T_s$, computing time T_1 shown in Fig. 7 is necessary for the processor, so result $\hat{V}_A(m)$ of the former sampling time is given to the PWM circuit during this sampling time $mT_s - (m+1)T_s$. Then if the result of the computation is $\hat{V}_{A0}(m)$ as shown by the line $X_1 - X_2$ in Fig. 7, so area S_A , which is the shortage of this sampling time, is compensated during next sampling time by $\hat{V}_A(m+1)$ given

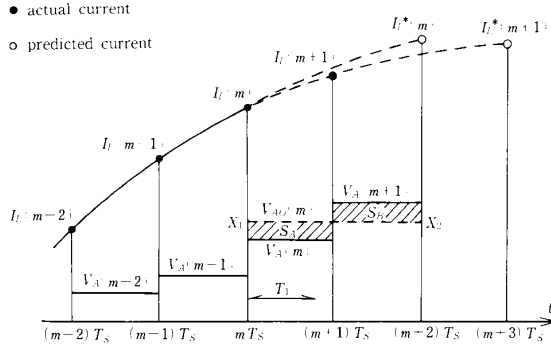


Fig. 7. Prediction of load current and compensation of computing time delay.

by (39), which compensates the shortage of S_A by S_B :

$$\hat{V}_A(m+1) = 2\hat{V}_{A0}(m) - \hat{V}_A(m). \quad (39)$$

As the target value of the condenser current, the q -axis component of after one discretization time $T = 2T_s$ should be given.

Another current target value $\hat{I}_C$ to correct voltage error is supplied from the voltage controller which operates, on the difference between voltage reference of one discretization time before $\hat{V}_C^*(k-1)$, and the condenser voltage at present $\hat{V}_C(k)$.

Thus, utilizing the prediction technique and discretization time of twice the sampling period, a short sampling time with little margin compared to computing time can be chosen. Using a high speed DSP, this sampling time can be shortened to around $100 \mu s$, so a high speed DDC that is suitable for a high speed switching device such as BiMOS can be realized. Compared to conventional methods, this is a remarkable feature.

In case of usual dead beat control systems, if $100 \mu s$ is necessary for computation, the sampling time should be chosen around $1 ms$ or more, so that not to decrease the utilization of the inverter voltage [4].

V. CONSTRUCTION OF TOTAL CONTROL SYSTEM

Combining the current minor loop and the voltage major loop of Sections III-D and E, and also the load current prediction and the compensation of computing time delay explained in Section IV, the total control system is constructed as shown in Fig. 6.

To transform the output voltage V_C , the inverter current I_A and the load current I_L to the d - q frame, three phase to d - q transformation blocks are provided. The current limiter is also provided, to limit the current reference to the current minor loop. After the block of compensation of computing time delay, the output voltage command signal to the PWM circuit is divided by the dc link voltage E_d , to compensate the influence of its change.

In case of the triangular wave comparison PWM, the d - q to three phase transformation block is necessary after the division block.

Elements of the coefficient matrices and the gain coef-

ficients of the controller obtained in Sections III-D and E can be given by the following equations using the (23):

$$\begin{aligned} a/c &= \omega L_S \cot \omega T \approx L_S/T = 1.95 \\ b/c &= \omega L_S = 0.189 \\ 1/c &= \omega L_S / \sin \omega T \approx L_S/T = 1.95 \\ a/e &= \omega C_P \cot \omega T \approx C_P/T = 0.78 \\ b/e &= \omega C_P = 0.075 \\ d/c &= f/e = \tan \omega T/2 \approx \omega T/2 = 0.048 \end{aligned} \quad (40)$$

where example of actual coefficients are obtained for typical circuit constants of simulation in Section VI. That is,

$$\omega = 2\omega f = 377 \text{ rad/s}$$

$$T = 128 \times 2 \mu s$$

$$L_S = 500 \mu H$$

$$C_P = 200 \mu F.$$

When ωT is sufficiently small as in this case, above approximation is valid.

By above approximation, the number of terms that change according to frequency decreases, which makes this controller easily applicable for adjustable frequency inverters.

VI. SIMULATION

Characteristics of this new principle of PWM inverter control have been studied using a hybrid computer (EAI2000 + Gould Sel 32/9705).

The inverter main circuit, the triangular wave comparator PWM circuit, and the load were simulated on the analog computer. The digital controller was simulated on the digital computer that simulates the 16-bit operation.

Delay of current and voltage sensors was simulated on the analog computer as a 2nd order ideal low pass filter of $20 \mu s$.

Characteristics of the system concerning the following parameters have been studied to confirm validity of this principle:

- relation between the sampling interval T_s and selection of the filter constants L_S and C_P
- influence of the switching time delay T_d , which is provided to avoid the conduction through the inverter
- tracking characteristics to the harmonic current source load
- adjustable frequency operation characteristics.

Besides usual wave forms, circular vector loci of the inverter voltage and current have been adopted to get better understanding of inverter operation. Typical results of simulation will be introduced below.

[Rating of Simulated Inverter]

9 kVA, 3 phase 173 V/100 V, 60 Hz (in case of constant frequency operation).

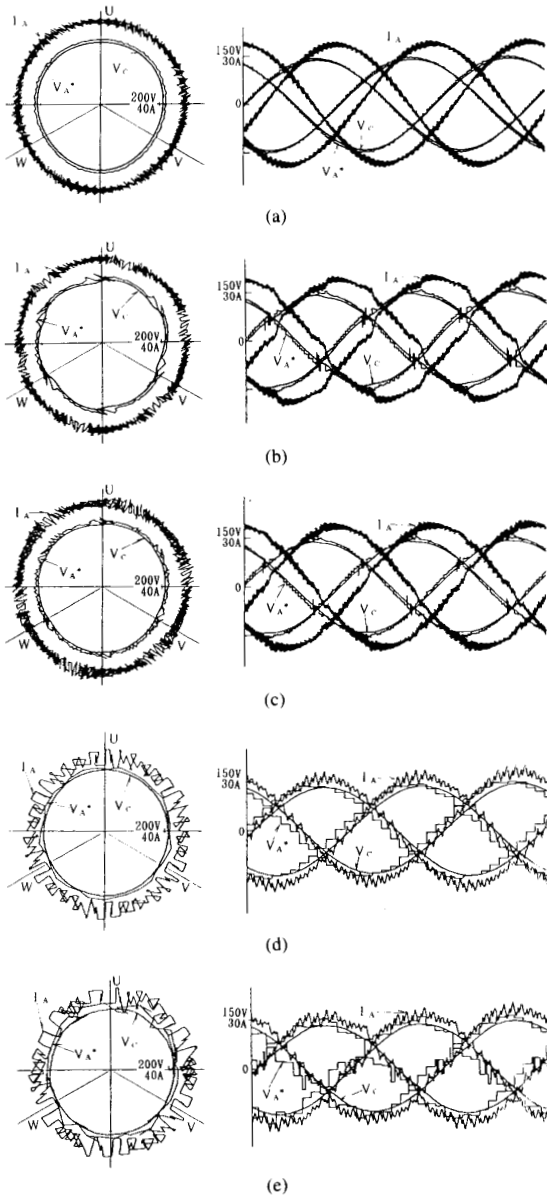


Fig. 8. Vector loci of inverter current and voltage. Linear L - R load 9 kVA. Output: 60 Hz, 173 V, 30 A. (a) Case A_1 , $T_S = 128 \mu s$, $T_d = 0$. (b) Case A_1 , $T_S = 128 \mu s$, $T_d = 3 \mu s$. (c) Case A_2 , $T_S = 80 \mu s$, $T_d = 3 \mu s$. (d) Case B, $T_S = 520 \mu s$, $T_d = 0$. (e) Case B, $T_S = 520 \mu s$, $T_d = 30 \mu s$.

Throughout the following simulations, the circuit condition of Case A_1 is mostly used. This condition corresponds to BiMOS inverters. In some cases, a condition of Case A_2 , which supposes a very high speed controller, is used.

The condition of CASE B corresponds to transistor inverters.

[Case A_1]

$L_S = 500 \mu H$, $C_P = 200 \mu F$ star, $T_S = 128 \mu s$, $f_{sw} = 15.6$ kHz (two triangular waves per one sampling time T_S).

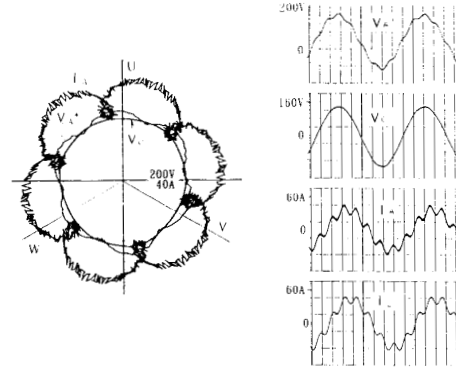


Fig. 9. Vector loci of inverter current response for load current of 7th harmonics. Condition: Case A_1 , $T_d = 0$, I_L ; fundamental = 30 A, 7th = 6 A (20 percent). Fundamental frequency = 60 Hz.

[Case A_2]

L_S and C_P ; ditto, $T_S = 80 \mu s$, $f_{sw} = 12.5$ kHz (one triangular wave per T_S).

[Case B]

$L_S = 2.5$ mH, $C_P = 500 \mu F$ star, $T_S = 520 \mu s$, $f_{sw} = 1.92$ kHz (one triangular wave per T_S).

1) Influence of sampling interval T_S and switching delay T_d .

Simulation wave forms and vector loci for a linear series L - R load are shown in Fig. 8.

Distortion of the output voltage V_C slightly increases when T_d of $3 \mu s$ is assumed for the BiMOS inverter, as shown in Fig. 8(a) and 8(b). However, if the sampling interval T_S is shortened from $128 \mu s$ to $80 \mu s$, the influence of T_d is suppressed very rapidly and the output voltage distortion becomes very small as shown in Fig. 8(c).

Signal V_A^* , which is the command value to the PWM circuit, shows operation of the digital controller to correct influence of T_d .

In case of the transistor inverter, the filter reactor L_S and the capacitor C_P are enlarged to cope with the long sampling interval ($T_S = 520 \mu s$). Then even in this case, fairly good sinusoidal output voltage V_C can be obtained as shown in Fig. 8(d) and 8(e).

2) Tracking characteristics to load harmonics

To examine influence of load harmonics, an ideal three phase current source load, which is composed of 100 percent fundamental component and 20 percent of seventh harmonic component, is connected as a load.

The result for the 7th harmonic shown in Fig. 9, indicates excellent response of this system.

3) Adjustable frequency operation

This system can be easily adopted as an adjustable frequency and voltage inverter for ac motor drives. In this case, the d - q frame for transformation, the output voltage and the condenser current reference should be changed according to frequency.

Simulation result of operation from 0 to 60 Hz is shown in Fig. 10.

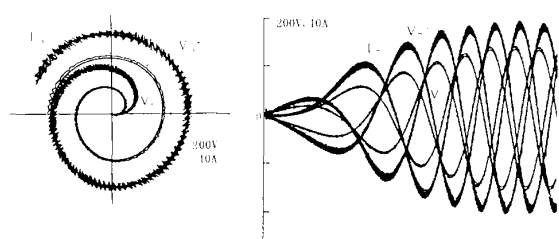


Fig. 10. Vector loci of inverter current and voltage for adjustable frequency operation. Condition: Case A₁, $T_d = 0$. Series L - R load, $0 \sim 60$ Hz.

VII. CONCLUSION

To construct microprocessor control system of the three phase sinusoidal voltage inverter, relation of the voltage and the current components of the three phase inverter was first clarified in discretized form on the d - q frame. Then using these relations, an algorithm of the dead beat current minor loop and the voltage major loop for sinusoidal voltage inverters has been developed.

A new method of compensation of computing time delay using load current prediction, which enables full utilization of the PWM period for the output voltage, has also been developed.

Validity of this control system has been proved by precise simulation using a hybrid computer.

After presentation of this paper in the 1987 Power Electronics Specialists Conference, several inverters using this principle have been manufactured and tested, and good experimental results coincide with the simulation have been obtained.

In these experiments, it has been studied that the circuit constants L_s and C_P should be selected within plus/minus 10 percent of the designed values to get good dead beat condition.

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